PARTICULAR SPECIFICATION P6B

APPENDIX P6B – Standard Control Features

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P6B.1 GENERAL CONTROL DESCRIPTION

The following sections describes standard control and monitoring features for TWRP including:

- Standard control modes;
- Protection, trips, interlocks and permissives;
- Standard PLC device control; and
- Instrumentation.

P6B.1.1 References

References used in this document

Document Number	Document Title	Remarks
WSW00-P-0014	P&ID LEGEND – LOCAL CONTROL STATION SHEET 1 OF 4	Motor driven equipment
WSW00-P-0015	P&ID LEGEND – LOCAL CONTROL STATION SHEET 2 OF 4	Motor driven equipment
WSW00-P-0016	P&ID LEGEND – LOCAL CONTROL STATION SHEET 3 OF 4	Actuated Valves (including penstocks)
WSW00-P-0017	P&ID LEGEND – LOCAL CONTROL STATION SHEET 4 OF 4	Chemical Dosing Pumps and Miscellaneous (including vendor package, analysers, chemical unloading)

P6B.2 STANDARD CONTROL MODES FOR INDIVIDUAL EQUIPMENT

P6B.2.1 Standard Equipment Modes

Most equipment such as motors and actuated valves will have control facilities on SCADA as well as in the field. The term ‘mode’ is used to denote the control method selected. Indication of the status of the device, and mode selected is shown next to the device on the graphic display in SCADA.

Motor-driven equipment and electrically actuated valves will generally have an associated Local Control Station (LCS), with the notable exception of some high voltage (HV) motors such as IPS pumps. In most cases, the LCS for actuated valves will be integral to the actuator.

Modes 1 and 2 cannot be selected or deselected from SCADA, because they are determined by a physical selector switch at the LCS.

Modes 3 to 5 can only be selected when the physical selector switch at the LCS is in the “SCADA” position. Selection of modes 3 to 5 is done from SCADA.

Each device has the following modes:

<u>Modes selectable from LCS Switch:</u>

1. **LCS** mode

   Device is controlled by an operator using pushbuttons or switches on a Local Control Station.

2. **OFF/ STOP** mode

   Mode selector switch at LCS is set to OFF (for equipment) or STOP (for motorised valves).

<u>Modes available for selection from SCADA if LCS switch set to “SCADA”:</u>

3. **AUTO** mode

   The device is controlled by the control logic (or sequences) programmed into the PLC.

   If while in this mode, the operator changes the selector switch at the LCS, the device will be forced to the mode selected on the switch.

   If while in this mode the device becomes “UNAVAILABLE’, the device will be forced to MANUAL mode. The operator may subsequently select it to OUT OF SERVICE mode if desired.

4. **MANUAL** mode

   The device is controlled by an operator from SCADA.

   When the mode selector switch at the LCS is switched to “SCADA”, a device will default to MANUAL mode. Other SCADA modes (AUTO or OUT OF SERVICE) may then be selected from SCADA.

   If while in this mode, the operator changes the selector switch at the LCS, the device will be forced to the mode selected on the switch.

5. **OUT OF SERVICE** mode

   This is for maintenance purposes (to replace a device or clean a tank, for example). All controls and alarms in the SCADA/PLC will be inhibited. Outputs are set to the safe position in this operating mode.

   If while in this mode, the operator changes the selector switch at the LCS, the device will be forced to the mode selected on the switch.

P6B.2.2 UNAVAILABLE Status (for Devices)

UNAVAILABLE is a status, not a mode. Electrically driven equipment will indicate UNAVAILABLE as a status on SCADA whenever the device is electrically isolated (but this shall not be used as evidence of isolation). The supply isolator position is monitored by the PLC to determine this condition. UNAVAILABLE is not a mode because it cannot be selected but rather it is just the result of a combination of conditions.

P6B.2.3 IPS Pump Control Modes

IPS Pumps are high-voltage VFD-driven pumps which will each have an LCP located within the relevant switchroom. Each LCP will have a PLC with i/o, and an independent touch-screen HMI. This pump PLC will directly monitor and control items specific to the respective pump, including wet-well level, pressures, and suction and discharge valves. In this way, any IPS Pump can be operated locally (from the switchroom), as an automatic unit, but independently of the master plant flow control sequence.

Thus, for IPS pumps, some of the standard control modes have a slightly different implementation as follows:

<u>Modes independent of Plant Master PLC</u>

- LCS Mode does not exist as the LCS consists only of an E-Stop function.
- Local PLC

  Control from LCP, using local HMI. Takes the place of 'LCS' mode. A semi-automatic mode in which a flow setpoint for the pump can be set, and in which start and stop sequences are automated.
- MCC / VFD

  Last resort control from VFD, requiring simultaneous manning of HPU control panel and pump operation from LCS at the motor level.

<u>Remote (plant master) PLC Control</u>

At the pump LCP HMI, Remote PLC control can be selected. The LCP HMI will then display "IN SCADA". The pump is under the overall control of the IPS plant master PLC in one of the three remaining modes:

- AUTO

  Plant master flow control automatically manages pump as part of the overall IPS.
- MANUAL

  Operator overrides plant master flow control to start and stop individual pumps from SCADA.
- OUT OF SERVICE (REMOTE)

  OUT OF SERVICE mode for IPS pump applies to individual pump only.

P6B.2.4 Special Cases

In addition to IPS Pumps, the list of modes in section P6B.2.1 omitted some other special cases. These are:

- Solenoid valves used for pneumatically actuated control valves. These may have mechanical override facilities requiring tools to operate. Therefore, field mechanical overrides for solenoid valves are not considered as normal local controls and these devices will not have a LCS mode as such.

P6B.3 DEVICE CONTROL FUNCTIONALITY

P6B.3.1 Location of Points of Control

Points of control in TWRP are specified in the table below.

Table P6A.3-1: Points of Control

Control Point	Operator Interface	Physical Location	Available Functions
SCADA	Operator Workstation	Control Room	Full control and monitoring if equipment is selected to "SCADA" at LCS
PLC	None	Process XLV Room	Automatic PLC Control
IPS Pump LCS	Local Control Station (LCS)	LCS at the field	Stop button Emergency Stop buttons Jog button Lamp Test button Local isolator
IPS Pump LCP	Touchscreen HMI	HV Switchroom	Full control and monitoring of individual pump as a unit.
MCC / VFD (not IPS Pump)	None	Switch room	No control from MCC/ VFD (fall back to Automatic PLC Control)
IPS Pump VFD	Physical or IOP controls	HV Switchroom	Mode Selector Switch "MCC / VFD" Start button Stop button Emergency Stop buttons Reset button
Blower	Touchscreen HMI LCP pushbuttons/ selector switches	Field or Blower Room	Full control and monitoring of individual equipment as a unit. Start button Stop button Emergency Stop buttons * Local / Remote selector switch Local isolator
	Touchscreen HMI	Master LCP in Switchroom or Process XLV Room	Setpoints System Alarms Local / SCADA selection
Centrifuge	Touchscreen HMI LCP pushbuttons/ selector switches	LCP in the Switchroom	Full control and monitoring of individual equipment as a unit. Start button Stop button Emergency Stop buttons * Local / Remote selector switch Local isolator
Fixed Speed Pump (not IPS Pump or Blower)	Local Control Station (LCS)	LCS at the field	Start button Stop button Emergency Stop buttons * LCS / OFF/ SCADA selector switch Local isolator (where applicable)
Motorised Valve Actuator	Integral controls	On the actuator	Mode Selector Switch "LCS/STOP/SCADA" (lockable in each position) Open button Close button Local isolator

Control Point	**Operator Interface**	**Physical Location**	**Available Functions**
Motorised Valve Actuator in inaccessible location	Detachable actuator control box located in operator accessible location	Near the actuator, in accessible location	Mode Selector Switch "LCS/STOP/SCADA" (lockable in each position) Open button Close button Open indicator lamp Close indicator lamp
Pneumatic Valve Actuator	None (refer to pneumatic control solenoids)	N/A	
Complex Machine other than blowers or centrifuges	Equipment Local Control Panel (LCP) – if provided by vendor	Field or switchroom	Vendor dependant
Vendor Package and Design and Build Package	System Local Control Panel (LCP) – if provided by vendor	Field or switchroom or process XLV room	Vendor dependant

* includes any additional emergency stop devices required by safety case e.g. conveyor lanyard switches

P6B.3.1.1 Mode Selection

AUTO and MANUAL mode selection is done at the SCADA operator workstation and is applied on an individual device (equipment/ actuated valve) basis. Group "all to AUTO" buttons may be provided if described in area-specific FDS documents.

The normal mode of operation is AUTO, with MANUAL selected only when required for special purposes.

When AUTO mode is selected, operator shall be required click a confirmation button in SCADA before auto mode will be activated.

When a PLC first starts-up, all devices will be in MANUAL mode; a feature to switch all devices within an area to AUTO mode (for certain operator access level) will also be available in SCADA.

Devices will be toggled to MANUAL by the PLC logic, whenever a Drive Trip (refer to section P6B.4.2) occurs. Resetting a drive trip (e.g. Overload trip) can only be done at the LCS, no remote reset is available at SCADA.

P6B.3.1.2 Action Upon Switch from AUTO to MANUAL Mode

A pump, blower, screen, or conveyor will stop when switched from AUTO to MANUAL mode.

A valve will maintain its current position (open/closed or hold at last position for modulating valve) when switched from AUTO to MANUAL.

P6B.3.1.3 Action Upon Switch from MANUAL to AUTO Mode

When switching from MANUAL to AUTO mode, the control settings of the device for the AUTO mode will become effective immediately if the device is employed for sequential operations.

P6B.3.1.4 Action Upon Switching a PID Controller Mode

The PID controller performs a bumpless transfer between device operating modes.

- From MANUAL to AUTO, controller output is to track the process value while the device is in MANUAL so there is no offset or error when switched to AUTO; and
- From AUTO to MANUAL, the manual output command is to track the controller's output while in AUTO such there is no bump in the output when switched to MANUAL.

P6B.3.1.5 Effect of Switching between SCADA (PLC) and LCS (FIELD) Modes

LCS/ SCADA mode selection is done from the LCS (for DOL motors and certain VFDs – refer to relevant LCS types in P&ID Legends), the Intelligent Operator Panel (IOP) on Variable Speed Drives (for IPS Pumps), the front panel selection on chemical dosing pumps or from the field (motorised valves and other equipment) by changing the selector switch from SCADA to LCS position.

P6B.3.1.5.1 SCADA to LCS Mode

Switching a device from SCADA to either MCC (for certain equipment only – refer to LCS types in P&ID legends) or LCS mode will stop motors.

For motorised valves (on/off), switching from SCADA to LCS mode will have no effect on valve position unless a specific action has been specified in the control description.

For modulating valves, switching from SCADA to LCS mode will hold the valve to the last position until the switchover is completed.

P6B.3.1.5.2 LCS to SCADA Mode

A controlled transfer from LCS or MCC (for certain equipment only – refer to LCS types in P&ID legends) to SCADA for motors is provided as follows:

- If the motor is running, it will be stopped because the default SCADA mode is MANUAL;
- If the motor is stopped, it should remain stopped; and
- The SCADA mode will initially be MANUAL.

The PLC configuration will match the valve status when switching a motorised valve from LCS to SCADA:

- If the valve is closed, it should remain closed;
- If the valve is open, it should remain opened;
- If it is a modulating valve, it should remain at the last position; and
- The SCADA mode will initially be MANUAL.

P6B.3.1.6 Out of Service Mode

The operator will switch a non-electrically driven device and chemical dosing pumps into OUT OF SERVICE (OOS) mode from MANUAL mode at the device’s faceplate on the SCADA.

From OUT OF SERVICE mode, a device can be switched by an operator action at the faceplate into MANUAL mode.

P6B.4 SAFETY PROTECTION, TRIPS, INTERLOCKS, AND PERMISSIVES

The distinction between personnel safety protection, drive trips, process interlocks, hardwired process interlocks and start permissives is described in the following sub-sections.

P6B.4.1 Personnel Safety Protection Signals

A personnel safety protection is a condition that stops the device in LCS (or MCC), AUTO and MANUAL (SCADA) modes because it operates directly in the electrical control circuit of the device e.g. E/Stop or pull cord or any other safety device.

These protections will cause a fault indication if a device was operating in AUTO or MANUAL (SCADA) modes because the control system will sense loss of the RUNNING feedback (and will possibly receive other more specific fault feedback).

The protection condition must itself be cleared, and, if the condition was device specific rather than a trip signal that affects multiple devices, then the operator must STOP the device (i.e. motor or motorised valve) and START the device again from the LCS (FIELD) before it can operate again.

Operation in LCS or MCC may will be possible as soon as the protection condition has been cleared (unless the condition is latched by a local safety relay or similar).

Logic for safety protections may be implement either as hardwire circuits or within safety rated PLC's.

When any safety device is activated for personnel safety protection, the safety device shall:

- Provides mechanical latching upon actuation (the device shall need manual physical re-arming to be healthy again);
- Be wired to a dedicated digital input channel in the motor control device (DOL, Soft Starter, VFD) and this will individually identifiable in the SCADA. However, serial wiring of safety devices is allowed (NC contacts);
- Upon activation, the motor control device shall STOP, regardless the status of the control sequence or command given by SCADA or LCS (FIELD);
- Upon activation, the electrical circuitry will remove control voltage to the LCS (FIELD). This will cause the drop of the REMOTE signal coming from the LCS;
- Control system will consequently change the AUTO status of the system to MANUAL. Therefore, the motor control device is not available for automatic sequence. Start command shall be removed;
- Motor control device faceplate will show: STOP, EMERGENCY STOP, MAN;
- Upon investigation and re-arming of the safety device, the aforementioned faceplate shall show: STOP, MAN. No further reset shall be required;
- Operator intervention will be required to make the motor control device available again for the automatic sequence (by switching MANUAL to AUTOMATIC in the SCADA);
- If available, the standby equipment shall start to ensure the right automatic process sequence;
- The event shall be registered in the alarm list with HIGH priority (subject to detailed alarm categorisation work).

P6B.4.2 Drive Trips

A drive trip is a fault condition that stops a device in both AUTO and MANUAL modes (and for IPS Pumps, Local PLC mode) via PLC control logic e.g. mechanical faults where further investigation or maintenance is required (e.g. excessive bearing temperature). The control system will also toggle the device mode to MANUAL (not for IPS Pumps).

The drive trip condition must itself have cleared (and been RESET if condition is one that latches), and the operator must switch the device mode back to AUTO (or use the START button to restart in MANUAL), in order to restart.

Some drive trips may be hardwired.

When drive trip happens, the tripped device shall:

- Provides logical and electrical latching upon actuation (to prevent unexpected start of the faulty equipment);
- Tripped signal shall be wired to a dedicated digital input channel in the motor control device (DOL, Soft Starter, VFD) and this will individually identifiable in the SCADA. However, serial wiring external monitoring devices is allowed (NC contacts) for multiple instances of similar condition (e.g. DE bearing temp and NDE bearing temp);
- Upon activation, the motor control device shall TRIP, regardless the status of the control sequence or command given by SCADA or LCS (FIELD);
- Upon activation, the electrical circuitry will remove control voltage to the LCS (FIELD). This will cause the drop of the REMOTE signal coming from the LCS;
- Device mode will change to MANUAL and the run command shall be removed;
- Upon investigation and clearing of the root cause, the operator will press the RESET button on the MCC cubicle. No further SCADA reset shall be required;
- Operator intervention will be required to make the motor control device available again for the automatic sequence (by switching MANUAL to AUTOMATIC in the SCADA);
- If available, the standby equipment shall start to ensure the right automatic process sequence;
- The event shall be registered in the alarm list with a MODERATE priority (subject to detailed alarm categorisation work).

P6B.4.3 Process Interlocks

A process interlock is a condition that stops the device in AUTO mode e.g. a valve must be open or a start level in a tank must be reached before a pump can start. The interlock condition does not require the operator to reset the device, which will restart when the interlock clears. It doesn't toggle the device out of AUTO mode either.

Process interlocks will always be implemented in software.

P6B.4.4 Hardwired Process Interlocks

Hardwired process interlock refers to any process interlocks which have been deemed necessary to be hardwired , as shown in P&IDs. These will operate in any mode.

Hardwired process interlocks shall:

- Have no latching of any type. Once the process conditions are cleared, the signal and system shall be shown as healthy;
- Be each wired to a dedicated digital input channel in the motor control device (DOL, Soft Starter, VFD) and the status of this input will be individually identifiable in the SCADA;
- Upon activation, the motor control device shall STOP, regardless the status of the control sequence or command given by SCADA or LCS;
- RUNNING feedback will be lost and control system will consequently change the mode to MANUAL and remove the Start command;

- Upon resuming normal process conditions, no further reset shall be required however the operator will be required to switch the device back to AUTO.

The event may be registered in the alarm list subject to detailed alarm categorisation work.

P6B.4.5 Start Permissives

A start permissive is a condition that prevents a device from starting in AUTO mode but does not stop the device once the device is running.

The start permissive condition does not require the operator to re-start the device and has no impact upon the mode.

Start permissives will always be implemented in software.

P6B.4.6 Interlock Applicability

Depending on the selected control mode, different categories of interlocks will be relevant as per the table below.

Table P6A.4-1: Impact of Control Mode on Interlocks

Selected Mode	Applicable Interlock Categories
AUTO	• Process Interlocks (Start and Run Permissives) • Personnel Safety Protection (hardwired interlocks)
MANUAL	• Process Interlocks (Start and Run Permissives) • Personnel Safety Protection (hardwired interlocks)
OUT OF SERVICE	• Personnel Safety Protection (hardwired interlocks) Equipment Unavailable for Operation (Auto/Manual) Used for Maintenance/ Equipment Replacement Purposes
LCS (or Local PLC Mode for IPS Pumps)	• Personnel Safety Protection (hardwired interlocks) • (For IPS Pumps, also Start and Run Permissives)
VFD mode for IPS Pumps	• Personnel Safety Protection (hardwired interlocks)

P6B.4.7 Display of Interlock Status on SCADA

The following information will be presented to an operator from the SCADA for each device (i.e.: a pump, a valve etc.):

- Drive Trips;
- Process Interlocks;
- Hardwired Process Interlocks (where applicable);
- Start Permissives; and
- Estop active.

P6B.5 PLC HARDWARE AND COMMUNICATION MONITORING

The following alarms will be generated in SCADA to alert the operator to failure in PLC hardware components and/ or communication signals:

- I/O module hardware failure alarm;
- Network switch failure alarm;
- Communications failure between PLC and RIO / MCC / VFDs;
- Peer-to-peer PLC communication failure alarm (watchdog).

When peer-to-peer PLC communication fails, each PLC shall generally retain its last operating status and remote setpoints in AUTO operation, unless this would be unsafe to personnel or would allow significant equipment damage to occur, in which case individual circumstances will be addressed in the specific PCN.

RIO and motor starters/VFD's shall STOP when communications to the controlling PLC are lost for a sustained period (more than a few seconds). Momentarily communication losses during fail-over of hot standby devices should not normally impact the final device.

P6B.6 PLC CONTROL OF STANDARD DEVICES

Standard devices are defined largely by function blocks and associated SCADA operation faceplates. They comprise of standard PLC functions, such as typical PID controllers, typical input and output channel blocks, duty/standby functions, math and logic functions, etc. as well as virtual images of physical devices in the field, such as actuated valves, motor drives and even package equipment, etc.

P6B.6.1 Standard Fixed Speed (DOL / Softstarter) Motor Control

P6B.6.1.1 Inputs/Outputs

A standard fixed speed motor has the following minimum PLC I/O signals (via data network connection):

Table P6A.6-1: Standard Fixed Speed Motor I/O Signals

PLC Digital Inputs:	PLC Digital Outputs:	PLC Analogue Inputs:	PLC Analogue Outputs:
Running	Run Command	NA	NA
Stopped			
Fault			
Available			
Power On			
E-Stop			
SCADA Mode Selected			
OFF Mode Selected			
LCS Mode Selected			

P6B.6.1.2 Device Statuses

The device has four statuses, RUNNING, STOPPED, UNAVAILABLE and FAULT, and will be displayed based on the inputs from the MCC and LCS. The status will be one of the following:

Table P6A.6-2: Standard Fixed Speed Motor Device Statuses

RUNNING SIGNAL	STOPPED SIGNAL	E-STOP SIGNAL	FAULT SIGNAL	AVAILABLE	POWER_ON SIGNAL	RESULTING STATUS
ON	OFF	OFF	OFF	ON	ON	RUNNING
OFF	ON	OFF	OFF	ON	ON	STOPPED
		ON	OFF	OFF	ON	UNAVAILABLE
			ON	ON	ON	FAULT
				OFF	ON	UNAVAILABLE
					OFF	UNAVAILABLE

P6B.6.1.3 Remote Operation

When in AUTO mode, the RUN COMMAND output is used to run the drive. The RUN COMMAND output is held ON for the entire duration the drive is required to run.

When in MANUAL mode, the RUN COMMAND output is used to run the drive when the START button is selected from the drive pop-up. The RUN COMMAND output is held ON until either the STOP button is selected from the drive pop-up or the device is stopped due to a fault, drive trip, safety protection or change of mode.

A failure signal due to start or stop fault is generated by the PLC when a run command has been issued and the running feedback has not been received within 5 seconds (nominal). This condition will behave in the same manner as a drive trip. The drive will be switched to MANUAL and a RESET from the SCADA system will be required before the drive can be returned to AUTO. The reset request will only be successful if any motor fault has already been reset at the MCC.

When OUT OF SERVICE (OOS) is selected from the SCADA, RUN COMMAND is blocked, as well as all alarms associated to the pump.

The number of starts and running hours is monitored by the PLC and displayed on the drive pop-up. All status changes are logged for trending and historical reporting.

P6B.6.1.4 LCS Operation

In LCS mode, the drive is controlled from the LCS at the field. The LCS/OFF/SCADA selector switch is selected to LCS then the START pushbutton is pressed at the LCS to start the drive. Once started, the RUNNING indicator is displayed on the LCS. A running drive may be stopped by pressing the STOP pushbutton at the LCS. If the device fails, the FAULT indicator will light at the LCS. No RESET button is available at the LCS, operator shall STOP and START the device at the LCS but must RESET drive faults at the MCC.

There is an E-STOP (emergency stop) button at the LCS. If the E-STOP button at LCS is pressed, the drive will be stopped and will not be restarted until the latching button is reset and a new run command is issued. This applies in all operational modes.

P6B.6.2 Standard Variable Speed Motor Control

P6B.6.2.1 Inputs/Outputs

A standard Variable Frequency Drive (VFD) motor has the following minimum PLC I/O signals:

Table P6A.6-3: Standard VFD Motor I/O Signals

PLC Digital Inputs:	PLC Digital Outputs:	PLC Analogue Inputs:	PLC Analogue Outputs:
Running	Run Command	Speed	Speed Control
Stopped		Power (kW)	
Fault		Current	
Available		Frequency	
Power On		Voltage	
E-Stop			
SCADA Mode Selected			
OFF Mode Selected			
LCS Mode Selected			

A potentiometer is provided at the LCS to allow speed adjustment at the field.

P6B.6.2.2 Device Statuses

The drive has four statuses, RUNNING, STOPPED, UNAVAILABLE and FAULT, and will be displayed based on the inputs from the MCC and LCS. The status will be one of the following:

Table P6A.6-4: Standard VFD Motor Device Statuses

RUNNING SIGNAL	STOPPED SIGNAL	E-STOP SIGNAL	FAULT SIGNAL	AVAILABLE	POWER_ON SIGNAL	RESULTING STATUS
ON	OFF	OFF	OFF	ON	ON	RUNNING
OFF	ON	OFF	OFF	ON	ON	STOPPED
		ON	OFF	OFF	ON	UNAVAILABLE
			ON	ON	ON	FAULT
				OFF	ON	UNAVAILABLE
					OFF	UNAVAILABLE

P6B.6.2.3 Remote Operation

When in AUTO mode, the RUN COMMAND output is used to run the drive. The RUN COMMAND output is held ON for the entire duration the drive is required to run. A SPEED CONTROL analogue output must be set by the PLC to control the speed that the drive will run at.

When in MANUAL mode, the RUN COMMAND output is used to run the drive when the START button is selected from the drive pop-up. The RUN COMMAND output is held ON until either the STOP button is selected from the drive pop-up or the device is stopped due to a fault, drive trip, safety protection or change of mode. The speed control signal for the pot on the LCS will be used.

A failure signal due to start or stop fault is generated by the PLC when a run command has been issued and the running feedback has not been received within 5 seconds (nominal). This condition will behave in the same manner as a drive trip. The drive will be switched to MANUAL and a RESET from the SCADA system will be required before the drive can be returned to AUTO. The reset request will only be successful if any motor fault has already been reset at the MCC.

When OUT OF SERVICE (OOS) is selected from the SCADA, RUN COMMAND is blocked, as well as all alarms associated to the pump.

The number of starts and running hours is monitored by the PLC and displayed on the drive pop-up. All status changes are logged for trending and historical reporting.

P6B.6.2.4 LCS Operation

In LCS mode, the drive is controlled from the LCS at the field. The LCS/OFF/SCADA selector switch is selected to LCS then the START pushbutton is pressed at the LCS to start the drive. Once started, the RUNNING indicator is displayed on the LCS. A running drive may be stopped by pressing the STOP pushbutton at the LCS. If the device fails, the FAULT indicator will light at the LCS. No RESET button is available at the LCS, operator shall STOP and START the device at the LCS but must RESET drive faults at the MCC.

There is an E-STOP (emergency stop) button at the LCS. If the E-STOP button at LCS is pressed, the drive will be stopped and will not be restarted until the latching button is reset and a new run command is issued. This applies in all operational modes.

P6B.6.3 Chemical Dosing Pump Control

P6B.6.3.1 General

Some chemical dosing pumps will be metering pumps with built-in VFD. The stroke length may be adjusted manually on the pump or set by a PLC command. The stroke rate (i.e. speed) is controlled by the PLC via Ethernet communication. The pumps can be selected to LCS (FIELD) or SCADA (REMOTE) mode on the pump itself or statuses will be displayed on the SCADA system.

P6B.6.3.2 Inputs/Outputs

A standard chemical dosing pump has the following minimum PLC I/O signals:

Table P6A.6-5: Standard Chemical Dosing Pump I/O Signals

Digital Input:	**Digital Output:**	**Analogue Input:**	**Analogue Output:**
Running	Run Command	Speed (Stroke Rate)	Speed (Stroke Rate Control)
Fault		Stroke Length	Stroke Length Control
Available		Power (kW)	
Power On			
E-Stopped			
SCADA Mode Selected (Available)			
OFF Mode Selected			
LCS Mode Selected			

P6B.6.3.3 Device Statuses

The drive has three statuses, RUNNING, STOPPED and FAULT, and will be displayed based on the E-STOP, RUNNING and FAULT inputs. The status will be one of the following:

Table P6A.6-6: Standard Chemical Dosing Pump Device Statuses

RUNNING SIGNAL	**E-STOP SIGNAL**	**FAULT SIGNAL**	**AVAILABLE**	**POWER ON SIGNAL**	**RESULTING STATUS**
ON	OFF	OFF	ON	ON	RUNNING
OFF	OFF	OFF	ON	ON	STOPPED
	ON	OFF	OFF	ON	UNAVAILABLE
		ON	ON	ON	FAULT
			OFF	ON	UNAVAILABLE
				OFF	UNAVAILABLE

P6B.6.3.4 Remote Operation

When in AUTO mode, the RUN COMMAND output is used to run the drive. The RUN COMMAND output is held ON for the entire duration the drive is required to run. A STROKE RATE CONTROL analogue output must be set by the PLC based on process conditions to control the stroke rate that the drive will run at.

When in MANUAL mode, the RUN COMMAND output is used to run the drive when the START button is selected from the drive pop-up. The RUN COMMAND output is held ON until either the STOP button is selected from the drive pop-up or the device is stopped due to a fault, drive trip, safety protection or change of mode. A STROKE RATE CONTROL analogue output must be set by the PLC based on based on the value selected by the operator on the drive pop-up.

A failure signal due to start or stop fault is generated by the PLC when a run command has been issued and the running feedback has not been received within 5 seconds (nominal). This condition will behave in the same manner as a drive trip. The drive will be switched to MANUAL and a RESET from the SCADA system will be required before the drive can be returned to AUTO. The reset request will only be successful if any motor fault has already been reset at the pump / MCC.

When OUT OF SERVICE (OOS) is selected from the SCADA, RUN COMMAND is blocked, as well as all alarms associated to the pump.

The number of starts and running hours is monitored by the PLC and displayed on the drive pop-up. All status changes are logged for trending and historical reporting.

P6B.6.3.5 LCS (LOCAL) Operation

In LCS (LOCAL) mode, the pump is controlled using its integral controls. The LCS / OFF/ SCADA selector switch is selected to LCS then the pump is started using the on-board controls. Once started, the RUNNING indicator is displayed on the SCADA.

There may be emergency stop facilities provided independently of the pump.

P6B.6.4 Standard Pneumatic Modulating Valve Control

P6B.6.4.1 Inputs/Outputs

A standard modulating valve has the following minimum PLC I/O signals:

Table P6A.6-7: Standard Pneumatic Modulating Valve I/O Signals

Digital Input:	Digital Output:	Analogue Input:	Analogue Output:
NA	NA	Position Feedback	Position Command

P6B.6.4.2 Control Modes

The valve modes consist of AUTO, MANUAL or OOS (OUT OF SERVICE), selected from the SCADA system. LOCAL control of the valve may be achieved by locally overriding the air supply to the valve. There is no means of detecting this from the PLC.

P6B.6.4.3 Device Statuses

The PLC input will be monitored to determine the status of the valve. The status of the valve will be displayed based on the POSITION FEEDBACK input. The status will be one of the following:

Table P6A.6-8: Standard Pneumatic Modulating Valve Device Statuses

STATUS	ANALOGUE FEEDBACK
OPEN	> 95%
CLOSED	< 5%
TRAVELLING	> 5% & < 95%

P6B.6.4.4 Remote Operation

When in AUTO mode, the POSITION COMMAND output is used to position the valve when required, based on various process conditions.

When in MANUAL mode, the POSITION COMMAND output is set by the PLC based on the value selected by the operator on the valve pop-up.

Pneumatic actuators will not generally generate faults that can be signalled directly. A position discrepancy fault is generated internally by the PLC when the valve position command has been issued and the position feedback is not received within 30 seconds (nominal). When this condition, the valve will be switched to MANUAL and a RESET from the SCADA system will be required before the valve can be returned to AUTO.

All status changes are logged for trending and historical reporting.

P6B.6.5 Standard Pneumatic On/Off Valve Control

P6B.6.5.1 Inputs/Outputs

A standard pneumatic on/off valve has the following minimum PLC I/O signals:

Table P6A.6-9: Standard Pneumatic On/Off Valve I/O Signals

Digital Input:	Digital Output:	Analogue Input:	Analogue Output:
Opened	Open Command	NA	NA
Closed			

P6B.6.5.2 Control Modes

The PLC inputs will be monitored to determine the status of the valve. The valve modes consist of AUTO, MANUAL or OOS (OUT OF SERVICE), selected from the SCADA system. FIELD (LOCAL) control of the valve may be achieved by locally overriding the air supply to the valve or manual control of the solenoid valves feeding air to the valve. There is no means of detecting this from the PLC.

P6B.6.5.3 Device Statuses

The status of the valve will be displayed based on the OPEN and CLOSED inputs. The status will be one of the following:

Table P6A.6-10: Standard Pneumatic On/Off Valve Device Statuses

STATUS	**OPEN**	**CLOSED**
OPEN	ON	OFF
CLOSED	OFF	ON
TRAVELLING	OFF	OFF

P6B.6.5.4 Remote Operation

When in AUTO mode, the OPEN COMMAND output is used to position the valve when required, based on various process conditions.

When in MANUAL mode, the OPEN COMMAND output is used to open the valve when the OPEN button is selected from the valve pop-up. The OPEN COMMAND output is held ON until either the CLOSE button is selected from the valve pop-up or the valve should be closed due to a fault or protection or change of mode.

Pneumatic actuators will not generally generate faults that can be signalled directly. equipment shall be re-started from the field.

Fail signal to indicate open/close discrepancy fault is generated internally by the PLC when the valve open command has been issued (turned ON) and the open feedback is not received within 30 seconds (nominal) or the valve open command has been removed (turned OFF) and the closed feedback is not received within 30 seconds (nominal).

In either case the valve will be switched to MANUAL and a RESET from the SCADA system will be required before the valve can be returned to AUTO.

All status changes are logged for trending and historical reporting.

P6B.6.6 Standard Motorised On/Off Valve or Penstock or Gate Control

P6B.6.6.1 Inputs/Outputs

A standard motorised on/off valve or penstock or gate has the following minimum PLC I/O signals:

Table P6A.6-11: Standard Motorised On/Off Valve I/O Signals

Digital Input:	Digital Output:	Analogue Input:	Analogue Output:
Actuator Alarm/ Fault	Open Command	NA	NA
Opened	Close Command		
Closed			
LCS Mode Selected			
Stop Mode Selected			
SCADA Mode Selected			

P6B.6.6.2 Control Modes

The PLC inputs will be monitored to determine both the mode and status of the valve. Modes as listed above AUTO, MANUAL, LCS or STOP (OOS) are determined by a combination of the mode selection from SCADA and the LCS input from the field.

- When the SCADA input is OFF, the mode will be displayed as LCS; and
- When the SCADA input is ON, the mode will be displayed as per the SCADA selection of AUTO or MANUAL.

P6B.6.6.3 Device Statuses

The status of the valve will be displayed based on the OPEN, CLOSED and FAULT inputs. The status will be one of the following:

Table P6A.6-12: Standard Motorised On/Off Valve Device Statuses

OPEN SIGNAL	CLOSED SIGNAL	FAULT SIGNAL	RESULTING STATUS
ON	OFF	OFF	OPEN
OFF	ON	OFF	CLOSED
OFF	OFF	OFF	TRAVELLING
		ON	FAULT

P6B.6.6.4 Remote Operation

When in AUTO mode, the OPEN COMMAND and CLOSE COMMAND outputs are used to open/close the valve when required, based on various process conditions. When the valve is required to be open, the OPEN COMMAND output is pulsed ON for a preconfigured duration. When the valve is required to be closed, the CLOSE COMMAND output is pulsed ON for the same duration. Feedback for both the OPEN and CLOSED states is directly from the device.

When in MANUAL mode, the OPEN COMMAND output is used to open the valve when the OPEN button is selected from the valve pop-up. The OPEN COMMAND output is pulsed ON for a preconfigured duration. When in MANUAL mode, the CLOSE COMMAND output is used to open the valve when the CLOSE button is selected from the valve pop-up. The CLOSE COMMAND output is pulsed ON for a preconfigured duration. Feedback for both the OPEN and CLOSED states is directly from the device.

When a PLC fault condition has occurred, a reset can be issued to the PLC from the SCADA system. The FAULT input from the valve can only be reset at the valve.

Fail signal from open/close discrepancy fault is generated internally by the PLC when the valve open command has been issued and the open feedback is not received within a preset time delay (hard coded in PLC) or the valve open command has been removed and the closed feedback is not received within a preset time delay.

In either case the valve will be switched to MANUAL and a RESET from the SCADA system will be required before the valve can be returned to AUTO.

All status changes are logged for trending and historical reporting.

P6B.6.6.5 LCS (LOCAL) Operation

In LCS (LOCAL) mode, the actuated valve/ penstock/ gate is controlled from the actuator LCS.

The SCADA/ STOP/ LCS selector switch is selected to LCS then the OPEN/CLOSE switch is operated to open or close the valve/ penstock/ gate.

When switched to STOP position, the valve/penstock/gate stops and remains at the last position. Its status shows OPEN, CLOSED or TRAVELLING.

P6B.6.7 Standard Motorised Control Valve or Penstock or Gate Control with Positioner

P6B.6.7.1 Inputs/Outputs

A standard motorised (modulating) control valve or penstock or gate with positioner has the following minimum PLC I/O signals:

Table P6A.6-13: Standard Motorised On/Off Valve I/O Signals

Digital Input:	**Digital Output:**	**Analogue Input:**	**Analogue Output:**
Actuator Alarm/ Fault	Stop Command	Position Feedback	Position Command
Opened			
Closed			
LCS Mode Selected			
Stop Mode Selected			
SCADA Mode Selected			

P6B.6.7.2 Control Modes

The PLC inputs will be monitored to determine both the mode and status of the valve. Modes as listed above AUTO, MANUAL, LCS or STOP (OOS) are determined by a combination of the mode selection from SCADA and the LCS input from the field.

- When the SCADA input is OFF, the mode will be displayed as LCS; and
- When the SCADA input is ON, the mode will be displayed as per the SCADA selection of AUTO or MANUAL.

P6B.6.7.3 Device Statuses

The status of the valve will be displayed based on the POSITION FEEDBACK (in % opening position) and FAULT inputs. The status will be one of the following:

Table P6A.6-14: Standard Motorised Control Valve Device with Positioner Statuses

POSITION FEEDBACK	FAULT SIGNAL	RESULTING STATUS
% OPENING POSITION	OFF	POSITION
	ON	FAULT

P6B.6.7.4 Remote Operation

When in AUTO mode, the POSITION COMMAND output is used to open/close the valve when required, for control based on various process conditions. POSITION COMMAND (percentage opening) is generated automatically based on the feedback from the flow controller. POSITION feedback is generated directly from the device.

When in MANUAL mode, the POSITION COMMAND (% opening) is entered by the operator manually from the valve pop-up in SCADA to command the valve to open at a certain % opening position (overriding the flow controller). POSITION (% opening) feedback is generated directly from the device.

When a PLC fault condition has occurred, a reset can be issued to the PLC from the SCADA system. The FAULT input from the valve can only be reset at the valve.

Fail signal from position discrepancy fault is generated internally by the PLC when the valve POSITION command has been issued and the % position opening feedback is not received within a preset time delay (hard coded in PLC).

In either case the valve will be switched to MANUAL and a RESET from the SCADA system will be required before the valve can be returned to AUTO.

All status changes are logged for trending and historical reporting.

P6B.6.7.5 LCS (LOCAL) Operation

In LCS (LOCAL) mode, the modulating control valve/ penstock/ gate with positioner is controlled from the actuator LCS.

The SCADA/ STOP/ LCS selector switch is selected to LCS then the OPEN/CLOSE switch is operated to open (fully open)/close (fully close) the valve/ penstock/ gate.

When switched to STOP position, the valve/penstock/gate stops and remains at the last (% opening) position.

P6B.6.8 Duty Selection Arbiter

A configurable Duty Arbiter is provided for all equipment sets where duty/assist/standby operation is required. The Duty Arbiter is provided for AUTO/MANUAL mode selection.

In AUTO mode, when a duty rotate command is received, duty/assist/standby selection will be performed by the PLC based on an algorithm as determined specifically for each particular equipment set, as described in section P6B.6.9.

In MANUAL mode, the operator can select duty/assist/standby for each unit in the equipment set. If a valid combination has been selected, they may then confirm the selection by pressing the SET DUTY button.

A duty rotate command may be initiated by the PLC based on specific conditions (i.e.: least running hour) for the equipment set. Alternatively, the operator may request a duty rotation by pressing the ROTATE DUTY button.

P6B.6.9 Automatic Duty Rotation

Automatic duty rotation is implemented as per one of the following rules for the different configurations of equipment other than instruments or valve trains:

Table P6A.6-15: Automatic Equipment Duty Rotation Configuration

CONFIGURATION	OPERATION	METHOD	DESCRIPTION
Lead/Standby Lead/Lead/Standby	Continuous	A	Changeover automatically when system run hours since last duty change exceeds X (Unique setpoint for each equipment item) hours. Use pattern sort.
Lead/Standby Lead/Lead/Standby Lead/Follow/Standby	Intermittent	B	Rotate duty automatically each time system stops. Use pattern sort. Duty rotation philosophy for different process systems may differ depending on the system requirements and as specified in the relevant PCN. The general approach will be to ensure units reach major service interval milestones with sufficient time in between each unit reaching the milestone, to allow major services to be completed one by one.
Lead/Follow/Follow/Standby Lead/Follow/Follow/Follow	Continuous	C	Rotate duty automatically whenever there is a change in which pumps are running. Duty rotate to be done separately for running pumps and stopped pumps. Rotation to be based on hours run since last stopped (for running pumps) – shortest run hours to go to Lead etc. or hours since last run (for stopped pumps) – longest run hours to start next etc. The result of this method is that each time a pump needs to be stopped or started, the pump to start will be the one that has been stopped for longest and the one to stop will be the one that has been running longest.

There will be no automatic duty rotation of redundant instruments or whole process trains. Rotation of these will be by operator manual selection.

P6B.6.10 PID Controller

For all PID control loops, the SCADA system will have the following:

- Indication of measured variable (PV), set-point (SP) and output (MV);
- Set-point adjustment;
- AUTO/MANUAL selection;
- Output adjustment when in MANUAL;
- PID Control Term adjustment;
- Setpoint Source Selection Internal/External;
- Scaling Output 0-100%; and
- Output Tracking.

P6B.7 PLC CONTROL OF SEQUENCES

Sequences are defined largely by function blocks and associated SCADA operation faceplates. They comprise of standard PLC functions, such as typical PID controllers, typical input and output channel blocks, duty/standby functions, math and logic functions, etc.

P6B.7.1 Modes for Sequences

Sequences shall have the following operator selectable modes:

- Auto;
- Manual; and
- Out of Service (OOS).

Out of service will normally only be selected when a system is completely out of service due to physical maintenance / breakdown etc. In this mode, sequence alarms and all alarms for the devices making up the sequence shall be disabled.

Some types of equipment will have additional operator selectable modes. These are:

- Online
- Offline

Offline is an Operator selected mode for a system but not a device. It refers to a system out of normal main production process, but still available to be in other auto sequence (e.g. membrane cleaning).

P6B.7.2 Availability of Process Trains

The terms “available” and “unavailable” may be applied to process trains (as opposed to individual device) to describe a status of the train which is derived from the particular condition or status of the various sub-items of the train.

P6B.8 MOST OPEN VALVE CONTROL

Most Open Valve (MOV) control logic may be specified in specific PCN’s to achieve equal but efficient flow regulation of aeration air. MOV is a technique developed to eliminate the excess power demand created by constant pressure regimes in aeration control. MOV is a modification of the underlying control loops. When operating properly MOV results in a system pressure equal to the minimum value to obtain the desired distribution and control. MOV tries to maintain valve opening at one of the tanks/ basins at the most open position possible at all times and throttle other valves as little as possible to meet the flow requirements.

When MOV is used for aeration control it will operate as follows:

- The flow control valve (FCV) for each aeration zone will be regulated by the flow-controller based on the flow measured upstream of each FCV, and according to a remote set-point determined by the ratio controller, dissolved oxygen controller, or feed-forward ammonia controller.
- One of the FCV's will be manually designated the MOV and when its controller output exceeds a certain trigger point the blower header pressure will be increased. The controllers for the other FCVs will then respond to maintain their flow setpoints. Blower pressure will be decreased if the MOV controller output falls below a second trigger point. The final values used for the two trigger points must be fine-tuned at commissioning but can be calculated once the precise piping and value characteristics are known.

P6B.9 INSTRUMENTATION

P6B.9.1 Dual Redundant Instruments

In some critical areas at the plant, dual redundant instruments are provided. This section describes the operation for redundant instruments.

P6B.9.1.1.1 Level Transmitters

Redundant level transmitters shall work in a duty/standby configuration.

Readings are dismissed automatically if the 4-20mA signal is out of range.

The operator selects one transmitter as duty when both/all transmitters are healthy; otherwise the system will select a default duty transmitter. Auto rotation of duty transmitter will be allowed in the PLC program in the event that the duty transmitter signals a fault condition via an out-of-range loop current. This logic also applies when PLC is reset.

Individual readings are displayed on the SCADA as standard configuration.

The selected duty transmitter is used for all controls.

When the difference between readings from two/three instruments exceeds the preset value for a time delay, an alarm is generated to the SCADA. By responding to alarm, the operator manually performs duty/standby changeover where appropriate. The operator can take either of the transmitters out of service via SCADA.

P6B.9.2 Failed Instruments

A failed instrument may be selected as OOS (OUT OF SERVICE) from the SCADA system. This will suppress all related alarms. The last value of the instrument will be held but sequences dependant on the signal will take a faulted state.

A supervisor may set a fixed (simulated) value to be entered so that calculations dependant on this instrument will continue to be performed. The indication “SIM” or similar will be displayed.

A list of instruments in SIM will be automatically maintained by the SCADA software.

P6B.9.3 Bypass Instruments

For process system that has few trains/ units, e.g.: screens/ MBRs/ conveyors etc; when any train/ unit is taken OUT OF SERVICE or UNAVAILABLE, PLC shall be programmed to bypass instrument(s) associated with that train/ unit.

P6B.9.4 Autosampler

Auto-sampler operation (START, STOP) shall be able to be controlled, and status (FAULT) monitored from the SCADA.

END OF SECTION